ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

# ROBOTICS FUNDAMENTALS

## PAPER 2 · PRACTICE B

**MARKS: 100** **TIME: 3 hours**

**NC(V) Programming and Robotics, Level 2. Paper 2 style (design-related), built around one case study: a line-following robot for a school competition. Section A: objective questions (30) · Section B: design brief, specifications and block diagram, sensor logic and calibration, flowchart, Arduino and block (Scratch) programming, 3D printing, testing and evaluation (70).**

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

## INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
6. Draw diagrams and flowcharts neatly, using a ruler. A non-programmable calculator may be used.
7. Write neatly and legibly.

## SECTION A

## QUESTION 1

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1–1.10) in the ANSWER BOOK.

**1.1** A simple line-following robot usually has at least … infrared sensors under its front.

A one
B two
C ten
D zero

**1.2** Compared with a white surface, a black line reflects … infrared light.

A more
B the same amount of
C less
D blue

**1.3** In a flowchart, a decision is shown by a …

A rectangle.
B diamond.
C oval.
D parallelogram.

**1.4** The start and end of a flowchart are shown by a/an …

A oval.
B diamond.
C rectangle.
D arrow.

**1.5** In block (Scratch-style) programming, a "forever" block makes the robot …

A run the blocks once.
B stop the program.
C wait for one second.
D repeat the blocks endlessly.

**1.6** The Arduino function digitalRead() returns …

A HIGH or LOW.
B a value from 0 to 1 023.
C text.
D the voltage in volts.

**1.7** A popular microcontroller board for school robots is the …

A Microsoft Excel.
B Arduino Uno.
C a USB stick.
D Windows 11.

**1.8** The filament most often used to 3D print a robot chassis is …

A PLA.
B glass.
C steel.
D wood.

**1.9** A test that checks whether the robot can follow a curved line is a …

A cost test.
B colour test.
C functional test.
D safety test.

**1.10** A motor-driver board is needed because …

A it makes a noise.
B it stores the program.
C it measures light.
D the controller's pins cannot supply enough current for motors.

(10 × 1) [10]

## QUESTION 2

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (2.1–2.10) in the ANSWER BOOK.

**2.1** An IR sensor gives a different reading over black than over white.

**2.2** If both line sensors see white, the robot should drive straight ahead.

**2.3** The arrows in a flowchart show the direction of flow.

**2.4** A program without loops will keep the robot following the line forever.

**2.5** The battery should be mounted as high as possible to stop the robot from tipping over.

**2.6** A robot only needs to be tested once, after it is finished.

**2.7** A 3D-printed part with 100% infill is lighter than the same part with 20% infill.

**2.8** Line sensors should be mounted close to the floor.

**2.9** A design brief describes the problem that must be solved.

**2.10** Calibration means adjusting the sensor threshold for the real surface and lighting.

(10 × 1) [10]

## QUESTION 3

Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (3.1–3.10) in the ANSWER BOOK.

| | COLUMN A | | COLUMN B |
|---|---|---|---|
| 3.1 | Short statement of the problem and what must be designed | A | chassis |
| 3.2 | Limits the design must keep to, such as cost and size | B | calibration |
| 3.3 | Early working model used to test an idea | C | design brief |
| 3.4 | Setting the black/white threshold of the sensors | D | test log |
| 3.5 | Steps to solve a problem, written before coding | E | algorithm |
| 3.6 | Programming by joining blocks on the screen | F | motor driver |
| 3.7 | Board that controls the speed and direction of motors | G | constraints |
| 3.8 | Plan of the steps and decisions drawn with symbols | H | flowchart |
| 3.9 | Frame that holds all the robot's parts together | I | prototype |
| 3.10 | Record of the results of every test | J | block programming |

(10 × 1) [10]

**TOTAL SECTION A: [30]**

## SECTION B

**CASE STUDY.** Your college enters a schools robotics competition. Your team must build a small robot that follows a black line on a white mat from start to finish as fast as possible, and stops at the black finish bar. The robot has two IR line sensors (left and right), a start button, an LED, two DC motors with a motor driver, and an Arduino Uno.

## QUESTION 4

**4.1** Write a short design brief for the robot. (2)

**4.2** Give FOUR specifications or constraints for the robot. (4)

**4.3** Draw a neat, labelled block diagram of the robot showing the inputs, the process and the outputs. (6)

**4.4** Explain why the two IR sensors are placed on either side of the line. (2)

**4.5** Choose a power supply for the robot (for example 4 × AA cells or a 9 V battery) and give a reason. (2)

**[16]**

## QUESTION 5

**5.1** Copy and complete the table below. For each combination of sensor readings, write what the robot must do. (8)

| Left sensor | Right sensor | Action of the robot |
|---|---|---|
| white | white | 5.1.1 |
| black | white | 5.1.2 |
| white | black | 5.1.3 |
| black | black | 5.1.4 |

**5.2** Explain what sensor calibration is and why it is needed. (2)

**5.3** During calibration an analogue IR sensor reads 200 over white and 800 over black. Calculate a suitable threshold value to tell black from white. (2)

**5.4** The robot leaves the line on sharp bends. Explain why this happens and give TWO ways to fix it. (4)

**[16]**

## QUESTION 6

**6.1** Draw a flowchart for the robot: wait for the start button, then keep following the line (using the four cases in Question 5.1) until both sensors see black, then stop and switch on the LED. (8)

**6.2** The left sensor is on pin 2 and the right sensor on pin 3; each reads HIGH over black. Write the Arduino loop() function that follows the line, using the functions forward(), turnLeft(), turnRight() and stopRobot(), which already exist. (8)

**6.3** Describe how the same line-following logic would be built with blocks in a Scratch-style program. Name the blocks you would use. (4)

**6.4** What is a function in a program, and why do programmers use functions such as turnLeft()? (2)

**[22]**

## QUESTION 7

**7.1** Describe, in order, FOUR steps to 3D print the robot's chassis. (4)

**7.2** Name TWO slicer settings that affect the strength of the printed chassis. (2)

**7.3** Draw up a test plan with FOUR tests for the robot and the expected result of EACH. (4)

**7.4** Give TWO safety checks to do before switching the robot on for the first time. (2)

**7.5** Suggest TWO improvements that could make the robot win the race. (2)

**7.6** In a test run the robot completes a 3 m track in 12 s. Calculate its average speed. (2)

**[16]**

**TOTAL SECTION B: [70]**

**TOTAL: 100**

**END OF QUESTION PAPER**